

CCP6224 Object Oriented Analysis and Design

Lab 10 – UML Behavioural Diagrams

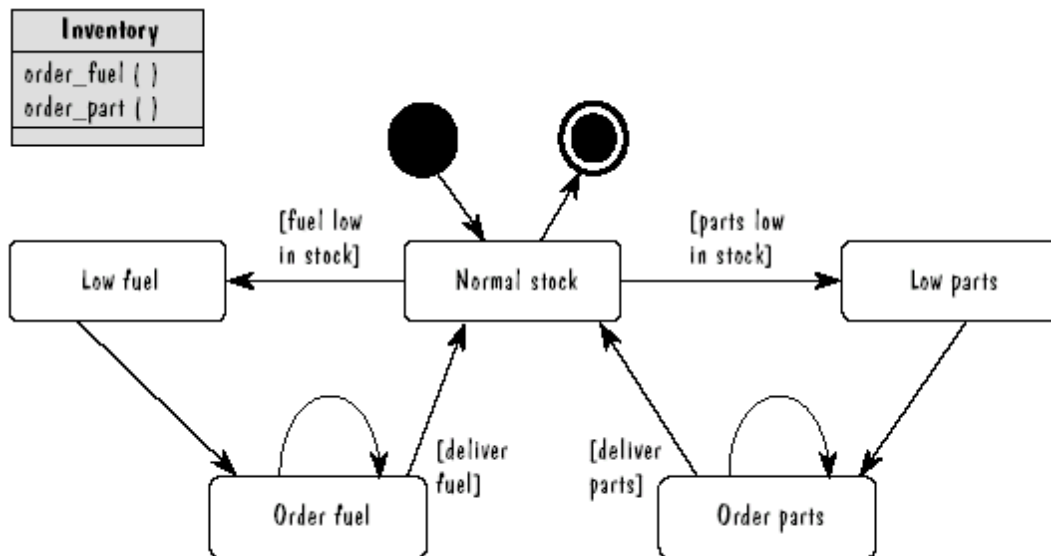
Lab outcomes

By the end of today's lab, you should be able to

- differentiate and draw different UML behavioural diagrams
- convert between diagrams

UML Behavioural Diagrams

1. Figure below indicates the State Diagram for the *inventory* software class of Royal Car Service Station:



- i) How many states exist in the inventory software?
- ii) What are the states?
- iii) What is a *guard*?
- iv) Under what conditions does the shop owner need to order more fuel or spare parts?
- v) When can the stock be back to normal?

2. Draw the state diagram for the following scenarios:

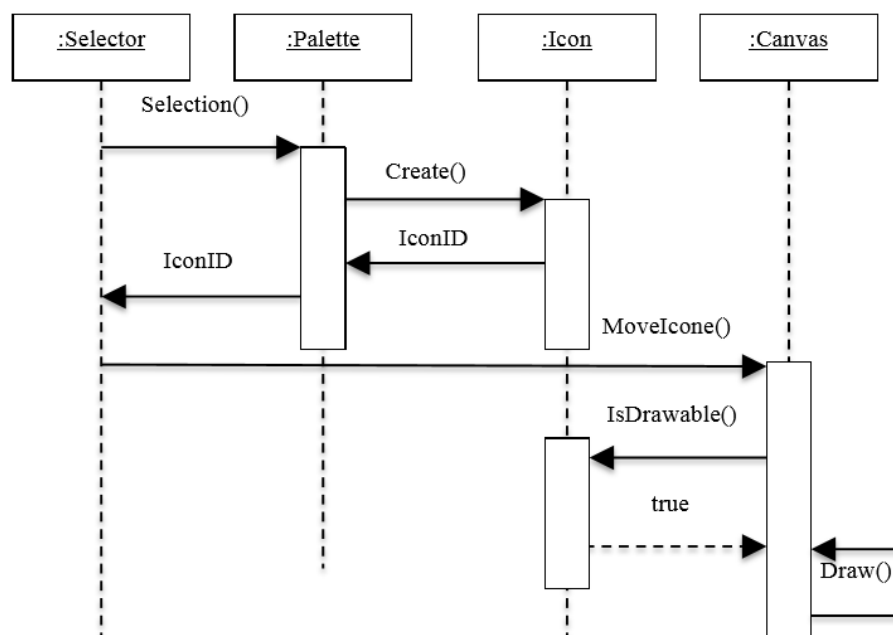
- a) A light bulb is connected to a push button switch. The light bulb is initially off. When the button is pushed, the bulb turns on but if the button is released it turns off
- b) A light bulb is connected to a push button switch. The light bulb is initially off. When the button is pushed, the bulb turns on. If the button is released it stays on but if the button is pushed again the bulb turns off and remains off until the switch is pushed again.

c) Imagine a lock that has a keypad with the digits 0 - 9. You need to enter the combination 5, 2, 4 and 7 to unlock the treasure. Any wrong number enter resets the lock into its initial state.

3. Draw the sequence diagram that illustrates the relationship between the customer and the shopping portal in the following scenario for online shopping.

- Customer searches the inventory for items – the portal returns results (if any to customer as a search result)
- Customers can opt to view details of search items or add items from the search result to the shopping cart
- Customers with shopping cart items can choose to checkout their items for payment when done

4. Convert the sequence diagram below to a collaboration diagram



5. Draw a swimlane diagram of the process of using an ATM card to get money from an ATM. The customer starts the use case by inserting their ATM card. The ATM then has to validate the card, and then ask the customer to enter the PIN. Then that PIN has to be checked by the bank to see if it's valid for that ATM card. If it is valid, then the customer can enter the amount of money they want to get. The bank then needs to check if there's enough money in the account to allow that amount to be taken out. If there is, then it should debit the account and let the ATM give cash to the user. When the user has taken the cash, the ATM should display the balance and eject the card.